

Homework 6: Ordination

S&DS 5360 | Multivariate Statistics

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Setup & Data Overview

We continue our work with the Healthy Minds Survey (HMS) data. For Correspondence ANalysis and NMDS, we will select related survey questions measured on the same scale. For these, we select:

- Depression (`phq9_1` to `phq9_9`): Nine questions on interest, mood, sleep, and energy levels over 2 weeks.
- Anxiety (`gad7_1` to `gad7_7`): Seven questions on nervousness, worrying, and restlessness over 2 weeks.

For Canonical Correspondence Analysis (CCA) and non-linear wireplots, we select continuous variables that may explain the variation in symptoms:

- Age (`age`): Continuous measure of respondent's age.
- Loneliness Score (`lonesc`): Sum of UCLA Loneliness Scale (3–9)
- Exercise Hours (`exerc`): Average hours per week spent exercising (1–5 scale).
- Internet Use (`internet_1`): Time spent online (1–8).
- Financial Stress (`fincur`): Stressfulness of current financial situation (1–5 scale).

We omit the code to preprocess the data.

Parts 1–4 | Correspondence Analysis

We begin by performing an indirect ordination to explore the underlying structure of mental health symptoms among students, particularly to reduce the dimensionality of our 16 symptom variables across PHQ-9 and GAD-7 to more easily visualize how they group together. To do so, we complete a Correspondence Analysis, which is ideal for our data because the symptom

data represents frequency counts (scales from 1 = “Not at All” to 4 = “Nearly Every Day”). CA maps both students and symptoms into low dimensional space via chi-square distances.

```
hms_ca <- cca(hms_symptoms)
summary(hms_ca)
```

Call:

```
cca(X = hms_symptoms)
```

Partitioning of scaled Chi-square:

	Inertia	Proportion
Total	0.1073	1
Unconstrained	0.1073	1

Eigenvalues, and their contribution to the scaled Chi-square

Importance of components:

	CA1	CA2	CA3	CA4	CA5	CA6
Eigenvalue	0.01769	0.01133	0.01063	0.008399	0.007819	0.007648
Proportion Explained	0.16484	0.10561	0.09905	0.078261	0.072859	0.071266
Cumulative Proportion	0.16484	0.27045	0.36950	0.447762	0.520621	0.591888

	CA7	CA8	CA9	CA10	CA11	CA12
Eigenvalue	0.006843	0.006118	0.005982	0.005109	0.004899	0.004552
Proportion Explained	0.063769	0.057013	0.055737	0.047603	0.045653	0.042412
Cumulative Proportion	0.655657	0.712670	0.768407	0.816010	0.861663	0.904075

	CA13	CA14	CA15
Eigenvalue	0.003909	0.003677	0.002708
Proportion Explained	0.036422	0.034268	0.025235
Cumulative Proportion	0.940497	0.974765	1.000000

We see that our total inertia is 0.1073, representing the total variance in our data (calculated as a scaled chi-square statistic). Axis 1 (CA1) explains 16.48% of the variance, while Axis 2 (CA2) explains 10.56%, or together, 27.04% of the variance in total.

Observe on our two-dimensional plot of Axes 1 and 2 that our clustered symptoms in blue text tightly overlap near the origin (0,0). Because there are so many students pulling the axes outward, the variation among symptoms themselves make them appear clumped in the middle. We provide a zoomed-in view on the right-hand side to more easily discern the symptoms.

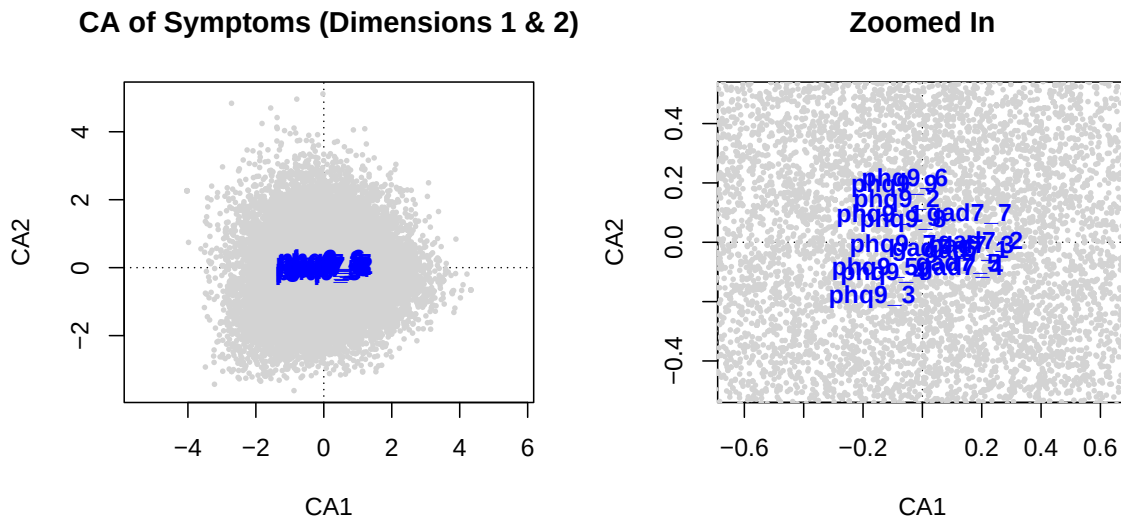
```

par(mfrow=c(1,2))

# Full Plot
plot(hms_ca, choices = c(1, 2), type = "n",
     main = "CA of Symptoms (Dimensions 1 & 2)")
points(hms_ca, display = "sites", col = "lightgrey", pch = 16, cex = 0.5)
text(hms_ca, display = "species", col = "blue", cex = 1, font = 2)

# Zoomed-In Plot
plot(hms_ca, choices = c(1, 2), type = "n",
     xlim = c(-.5, .5), ylim = c(-.5, .5),
     main = "Zoomed In")
points(hms_ca, display = "sites", col = "lightgrey", pch = 16, cex = 0.5)
text(hms_ca, display = "species", col = "blue", cex = 1, font = 2)

```



Upon visualizing the first two dimensions of the Correspondence Analysis, we see there is no clear evidence of an arch effect (data snaking). Rather than forming a classic U or V crescent shape, the data forms a dense, continuous cloud centered around the origin, with the symptom variables tightly clustered in the middle.

As discussed in class, this lack of snaking would be expected when there is complete species turnover along an environmental gradient. Our mental health symptoms do not behave this way; rather than “turning over,” symptoms are highly correlated and additive. For example, a student experiencing severe depression does not replace their more mild symptoms (like fatigue) with more severe ones (like self harm). Rather, they may experience both simultaneously. Since

these symptoms co-occur, the CA maps them as the tightly inter-correlated cluster we see in the plots.

Given the above, we conclude that mental health symptoms measured by PHQ-9 and GAD-7 are highly comorbid and do not represent distinct, mutually exclusive conditions among the student population. The lack of data snaking and the tight clustering of symptom variables near the origin indicate that students do not experience a turnover symptoms. Instead, the first two axes, which capture $\approx 27\%$ of the variance, likely represent a single, dominant gradient of overall psychological distress. Students scale along this gradient additively, experiencing these anxiety and depression symptoms simultaneously rather than in isolation of each other.

Parts 5–7 | Multidimensional Scaling

Now, we perform Non-Metric Multidimensional Scaling (NMDS) in one, two, and three dimensions. Unlike Correspondence Analysis, which relies on chi-square distances, NMDS's rank-based approach maps respondents into a low-dimensional space such that the distances between points optimally preserve the rank-order of their actual dissimilarities in the full 16-dimensional symptom space.

We elect to use the Bray-Curtis distance metric (as opposed to Euclidean distance) because of its robustness to joint absences. For example, if two students both select “not at all” for a number of severe symptoms, Bray-Curtis does not treat this shared absence as a strong indicator of similarity, focusing instead on the actually experienced symptoms. This is optimal for survey data like ours.

```
set.seed(4747)

df_subset <- df %>% sample_n(500)

hms_symptoms_subset <- df_subset %>% select(all_of(symptoms))
hms_env_subset <- df_subset %>% select(all_of(environment))

hms_nmds1 <- metaMDS(hms_symptoms_subset, distance = "bray", k = 1, trymax = 50)
hms_nmds2 <- metaMDS(hms_symptoms_subset, distance = "bray", k = 2, trymax = 50)
hms_nmds3 <- metaMDS(hms_symptoms_subset, distance = "bray", k = 3, trymax = 50)
```

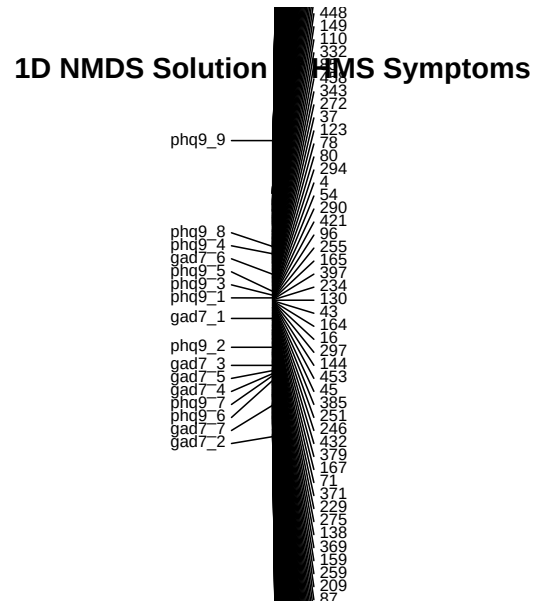
Note that we manually set `trymax = 50` to ensure the algorithm incorporates enough random starts to reliably find the global optimum within a reasonable number of runs. Furthermore, we sample 500 rows from our data, as our dataset containing 33,100 rows of symptom data is computationally expensive and unnecessary for ordination.

```

par(mfrow = c(1,1))

# 1D
plot(hms_nmds1, main = "1D NMDS Solution of HMS Symptoms")

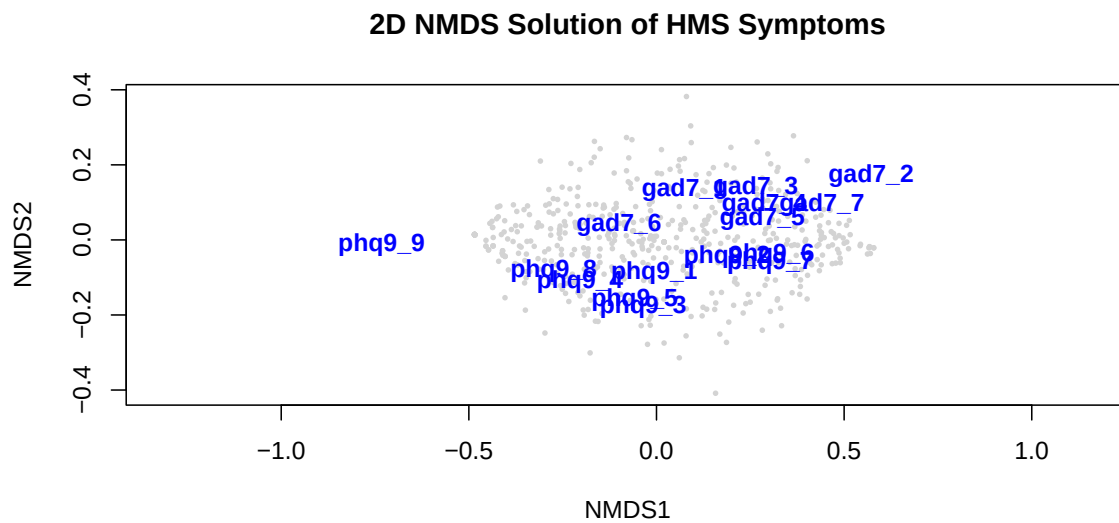
```



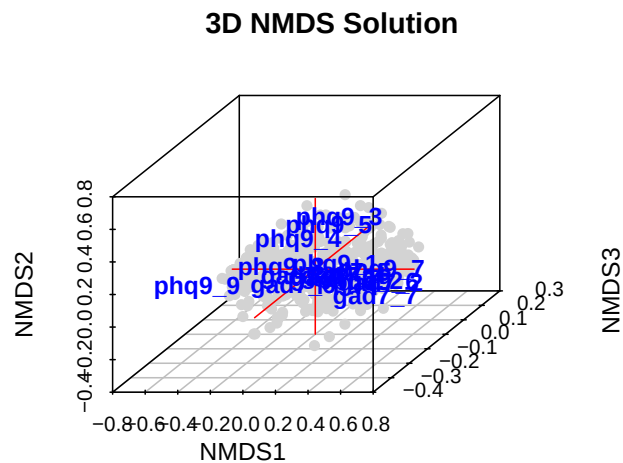
```

# 2D
plot(hms_nmds2, type = "n", main = "2D NMDS Solution of HMS Symptoms")
points(hms_nmds2, display = "sites", col = "lightgrey", pch = 16, cex = 0.5)
text(hms_nmds2, display = "species", col = "blue", cex = 1, font = 2)

```



```
# 3D
pl <- ordiplot3d(hms_nmds3, display = "sites", pch = 16, col = "lightgrey",
  main = "3D NMDS Solution")
sp <- scores(hms_nmds3, choices = 1:3, display = "species")
text(pl$xyz.convert(sp), rownames(sp), col = "blue", font = 2, cex = 1)
```



In the above plots, we see: - **1D NMDS:** The 1-dimensional plot forces all symptom variance onto a single axis, resulting in an overcrowded plot from which it is difficult to discern meaning-

ful relationships between individual symptoms. - **2D NMDS:** The 2-dimensional plot reveals a much clearer mapping of the symptom space. Observe the extreme observation `phq9_9` on the far left-hand side; this question specifically measures suicidal ideation and self-harm. The remaining general anxiety and depression symptoms form a looser cluster on the right. The visual separation makes sense: self harm is distinctly different from other, less severe depressive symptoms and requires acute care. A less drastic, but still notable, distinction is the vertical axis, which separates symptoms by clinical nature. Note that `phq_3` (sleep), `phq_4` (fatigue), and `phq_5` (appetite) cluster together at a distinctly lower elevation on the second axis. These questions represent the physical manifestations of depression. In contrast, the items situated higher on the y-axis are predominantly GAD-7 anxiety variables, such as uncontrollable worry and feeling on edge. This indicates that the NMDS algorithm successfully captured the latent structural divide between physical exhaustion and depressive symptoms compared to anxious states among the student population. - **3D NMDS:** The 3-dimensional plot adds an additional axis for depth which is not immediately helpful. While it helps pull apart the main cluster of symptoms that overlap in 2D, the overall topology remains the same, with `phq9_9` retaining the same distinct separation.

```
# Print stress numbers
stress_1d <- hms_nmds1$stress
stress_2d <- hms_nmds2$stress
stress_3d <- hms_nmds3$stress

cat("Stress for 1D:", stress_1d, "\n")
```

Stress for 1D: 0.2000061

```
cat("Stress for 2D:", stress_2d, "\n")
```

Stress for 2D: 0.1346835

```
cat("Stress for 3D:", stress_3d, "\n")
```

Stress for 3D: 0.1057252

```
# Plot
dimensions <- c(1, 2, 3)
stress_values <- c(stress_1d, stress_2d, stress_3d)

plot(dimensions, stress_values, type = "b", pch = 19, col = "blue", lwd = 2,
      xlab = "Number of Dimensions (k)", ylab = "Stress",
```

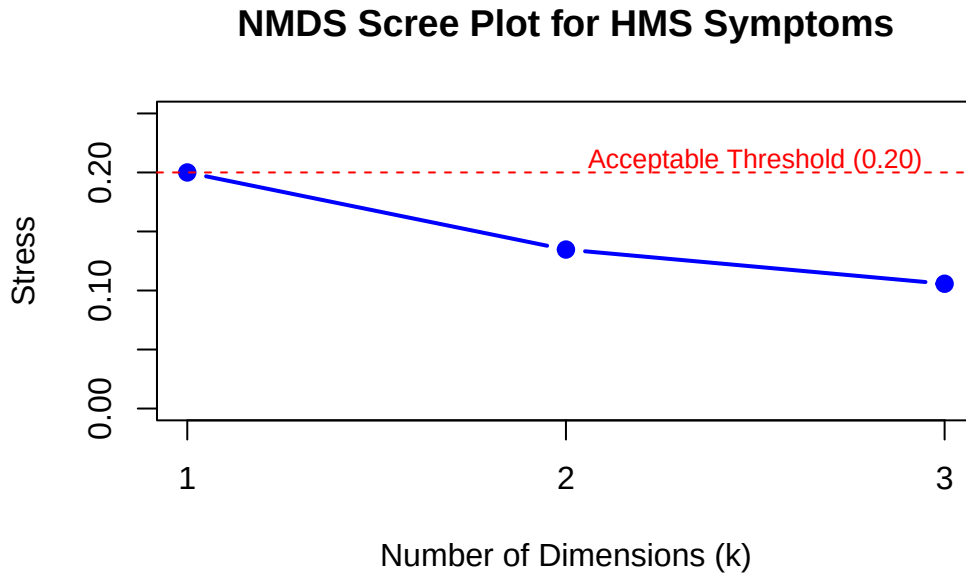
```

main = "NMDS Scree Plot for HMS Symptoms",
ylim = c(0, max(stress_values) + 0.05), xaxt = "n")

axis(1, at = c(1, 2, 3))

abline(h = 0.20, col = "red", lty = 2)
text(2.5, 0.21, "Acceptable Threshold (0.20)", col = "red", cex = 0.8)

```



The stress of our 1D, 2D, and 3D NMDS solutions were 0.2000, 0.1347, and 0.1057, respectively. As visualized in the scree plot, moving from a 1D to 2D solution provides a notable and steep reduction in stress, falling below the acceptable range of 0.20. This means that the 2D plot provides a reliable geometric representation of the symptom distances without severe distortion. While the 3D solution reduces the stress further, the 2D solution is optimal for this analysis as it perfectly balances an acceptable goodness-of-fit with visual interpretability, capturing the primary structural divides in the data. This confirms what we found in our visual analysis earlier.

Part 8 | Overlay of Continuous Variables

Proceeding with our 2-dimensional NMDS solution, we overlay the five continuous variables we selected at the start: age, loneliness score, exercise hours, internet use, and financial stress.


```
fit_env <- envfit(hms_nmds2, hms_env_subset, permutations = 999)
fit_env
```

***VECTORS

	NMDS1	NMDS2	r2	Pr(>r)	
age	-0.68500	0.72854	0.0074	0.137	
lonesc	0.91486	-0.40378	0.3228	0.001	***
exerc	-0.37569	0.92674	0.0144	0.029	*
internet_1	0.83691	-0.54733	0.0554	0.001	***
fincur	-0.98544	-0.17004	0.1595	0.001	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

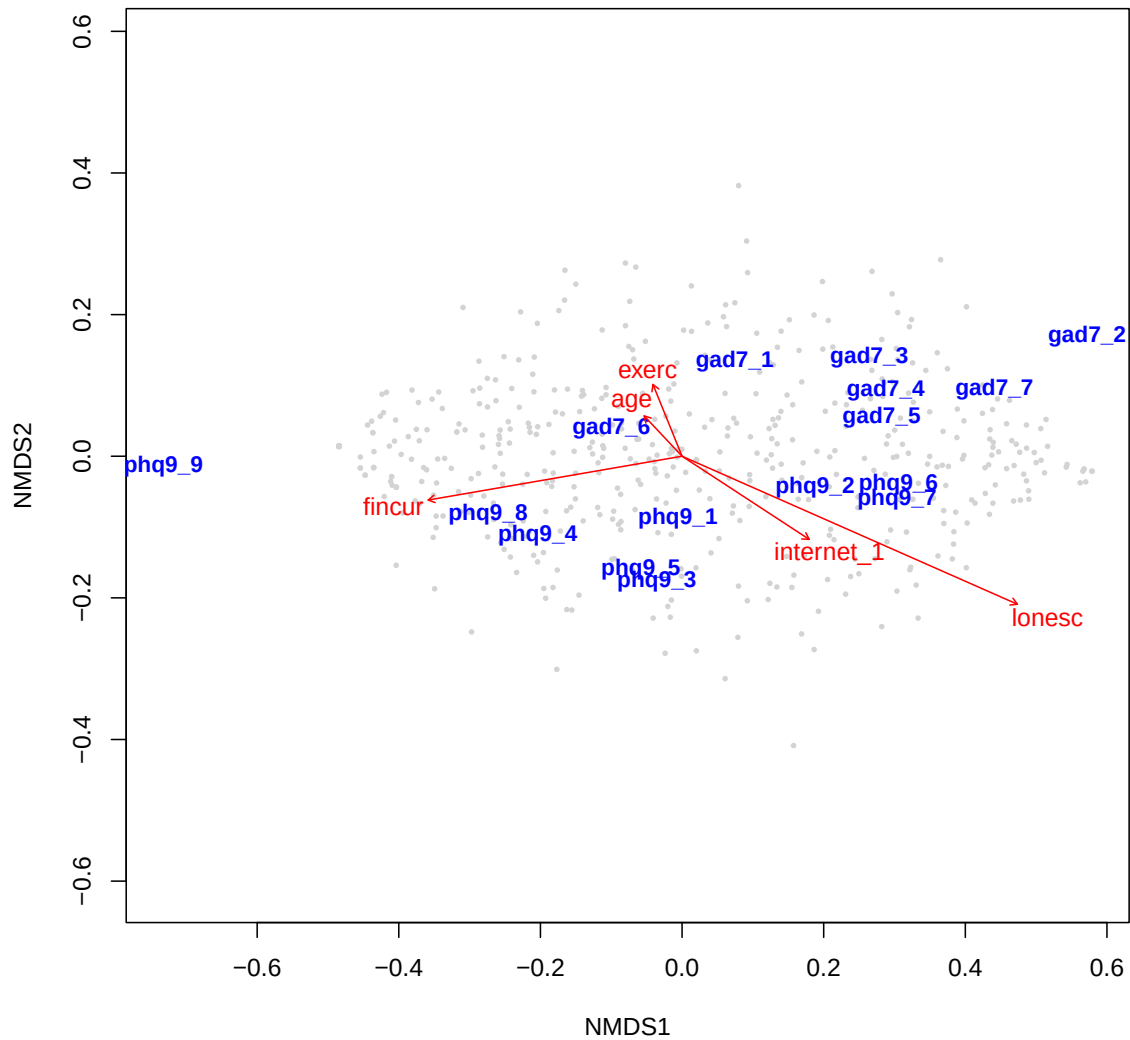
Permutation: free

Number of permutations: 999

Immediately notice that four of our five variables are statistically significant: **lonesc**, **exerc**, **internet_1**, and **fincur** (just not **age**). Looking at the corresponding R-squared values, observe that loneliness has an extremely high value of 0.3228, followed in magnitude by **fincur**, **internet1**, and finally, **exercise**. This tells us that loneliness, particularly, is a huge driver of where students land in the symptom space.

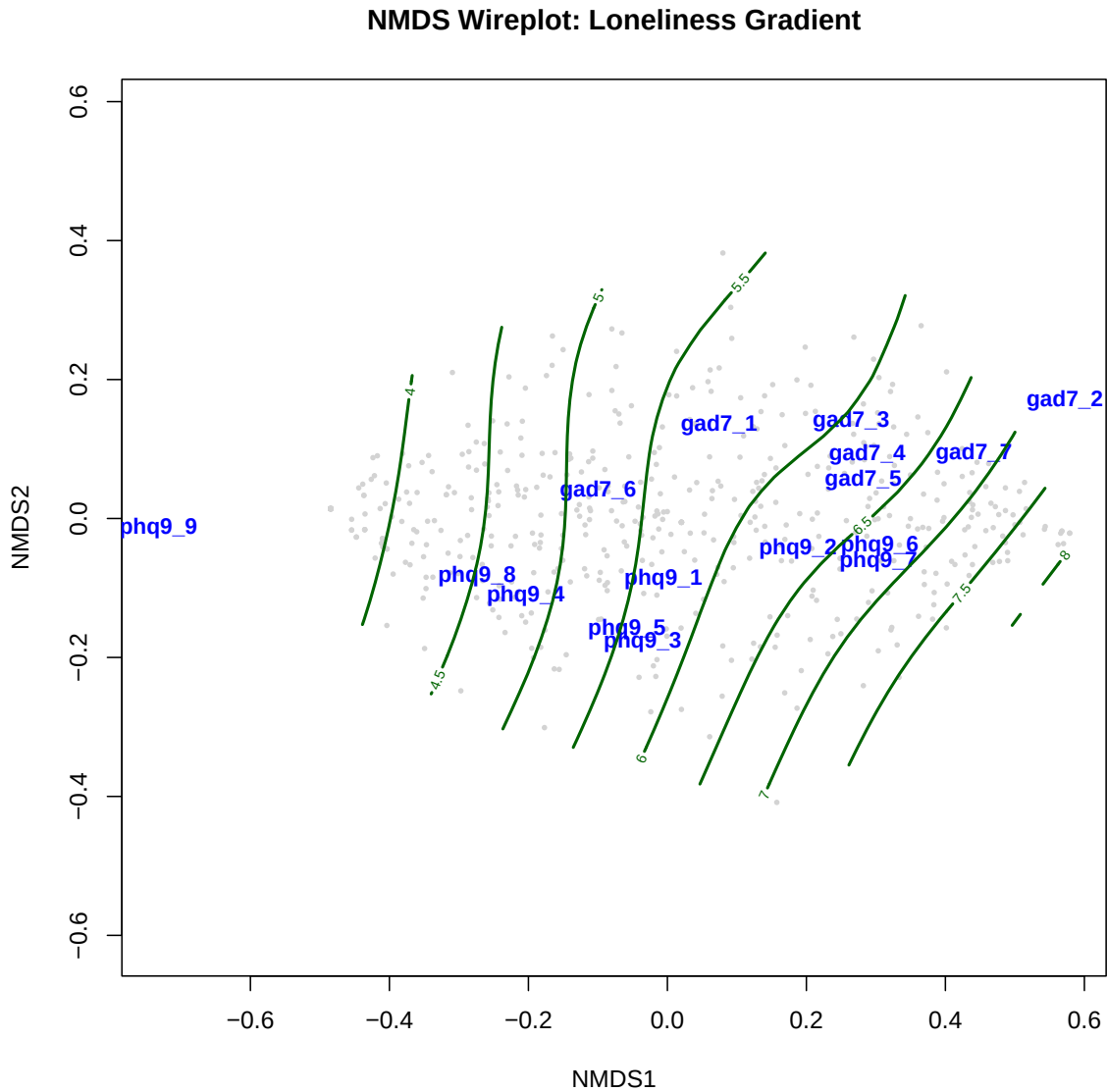
```
plot(hms_nmds2, type = "n", main = "NMDS with Environmental Vectors")
points(hms_nmds2, display = "sites", col = "lightgrey", pch = 16, cex = 0.5)
text(hms_nmds2, display = "species", col = "blue", cex = 0.9, font = 2)
plot(fit_env, col = "red", lwd = 2)
```

NMDS with Environmental Vectors



A visual plot in two dimensions confirms our numerical analysis . We note that the length of the vectors for each continuous variable match the magnitude of the aforementioned correlations. Now, observing the direction of each arrow, we see that **lonesc** and **internet_1** point down and to the right, which is where a blend of anxiety and depression symptoms lie. Meanwhile, **fincur** pulls hard to the left, acting along the primary axis where the severe outlier **phq9_9** sits. This seems to indicate that while loneliness is a major driver behind anxiety and depression, financial stress, in particular, seems to be a major driver behind more severe suicidal ideation.

```
plot(hms_nmds2, type = "n", main = "NMDS Wireplot: Loneliness Gradient")
points(hms_nmds2, display = "sites", col = "lightgrey", pch = 16, cex = 0.5)
text(hms_nmds2, display = "species", col = "blue", cex = 0.9, font = 2)
surf_lone <- ordisurf(hms_nmds2, hms_env_subset$lonesc, add = TRUE,
                      col = "darkgreen", lwd = 2)
```

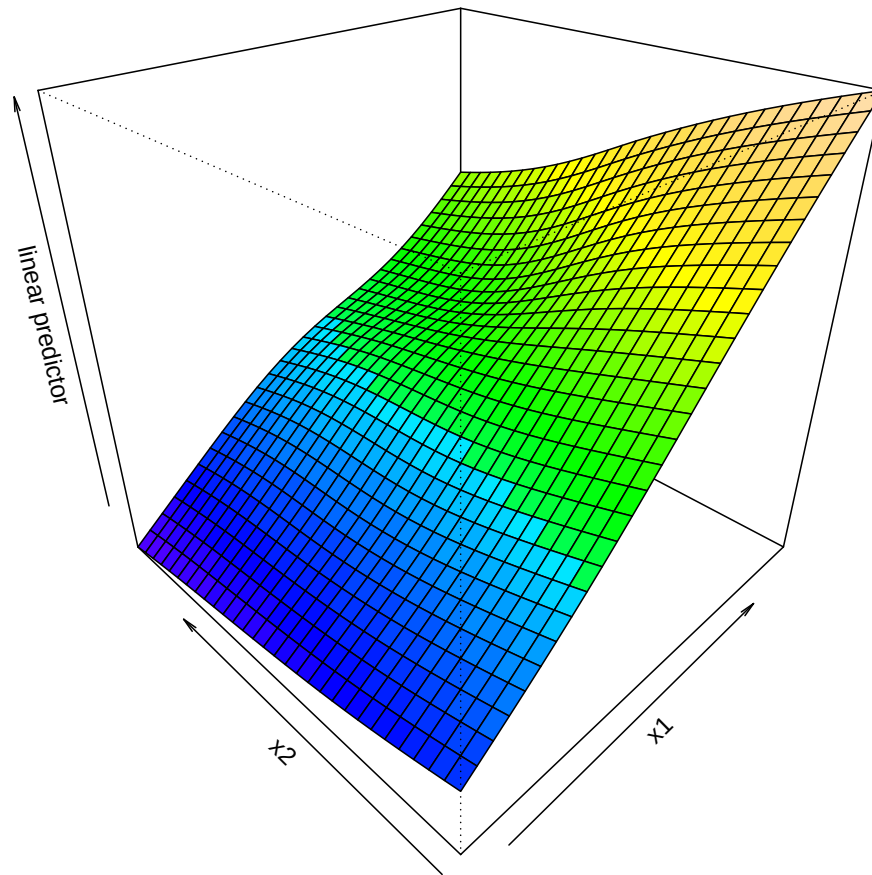


We further study loneliness given its large correlation of 0.3228 by generating a 2-dimensional wireplot. Recall that the green lines represent average loneliness scores, which generally in-

crease from left to right as we shift towards the dense cluster of anxiety and depression symptoms on the right.

```
vis.gam(surf_lone, main = "3D Surface of Loneliness over NMDS Space",  
        theta = -45, phi = 30, color = "topo")
```

3D Surface of Loneliness over NMDS Space



Now, observe the 3D surface plot of loneliness over NMDS space. The highest elevation (in yellow and orange) corresponds to highest loneliness scores, confirming that extreme loneliness maps into the clustered symptom space as we have seen previously.

Part 9 | Canonical Correspondence Analysis (CCA)

Finally, we use loneliness, financial stress, and internet use (our strongest correlated variables) to constrain a Canonical Correspondence Analysis (CCA) model.

```
# Model
hms_cca <- cca(hms_symptoms_subset ~ lonesc + fincur + internet_1,
              data = hms_env_subset)
summary(hms_cca)
```

Call:

```
cca(formula = hms_symptoms_subset ~ lonesc + fincur + internet_1,      data = hms_env_subset)
```

Partitioning of scaled Chi-square:

	Inertia	Proportion
Total	0.101787	1.00000
Constrained	0.001417	0.01393
Unconstrained	0.100369	0.98607

Eigenvalues, and their contribution to the scaled Chi-square

Importance of components:

	CCA1	CCA2	CCA3	CA1	CA2	CA3
Eigenvalue	0.000639	0.0005078	0.0002707	0.01466	0.01186	0.01006
Proportion Explained	0.006278	0.0049892	0.0026592	0.14405	0.11654	0.09879
Cumulative Proportion	0.006278	0.0112670	0.0139262	0.15798	0.27452	0.37331
	CA4	CA5	CA6	CA7	CA8	CA9
Eigenvalue	0.008298	0.008005	0.007183	0.006966	0.006031	0.005563
Proportion Explained	0.081527	0.078644	0.070572	0.068436	0.059255	0.054654
Cumulative Proportion	0.454837	0.533481	0.604053	0.672488	0.731743	0.786397
	CA10	CA11	CA12	CA13	CA14	CA15
Eigenvalue	0.004799	0.004527	0.004004	0.003454	0.002771	0.002186
Proportion Explained	0.047143	0.044478	0.039340	0.033933	0.027228	0.021480
Cumulative Proportion	0.833540	0.878019	0.917358	0.951292	0.978520	1.000000

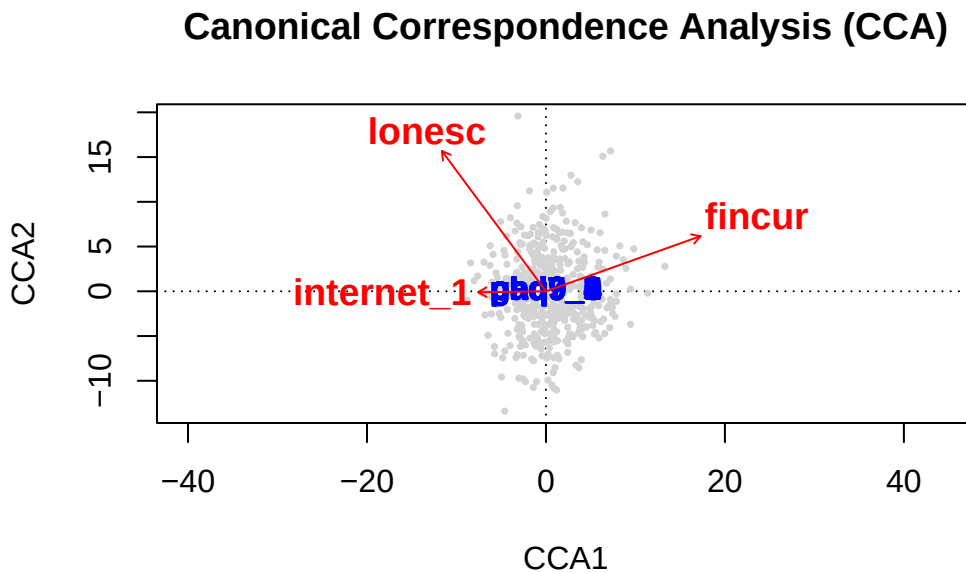
Accumulated constrained eigenvalues

Importance of components:

	CCA1	CCA2	CCA3
Eigenvalue	0.000639	0.0005078	0.0002707
Proportion Explained	0.450790	0.3582632	0.1909467
Cumulative Proportion	0.450790	0.8090533	1.0000000

Immediately observe that our total inertia is 0.1018, but our constrained inertia (the amount of variance explained by our selected continuous variables) is only 0.0014, or a mere 1.39%. We produce a triplot to analyze:

```
# Plot
plot(hms_cca, type = "n", main = "Canonical Correspondence Analysis (CCA)")
points(hms_cca, display = "sites", col = "lightgrey", pch = 16, cex = 0.5)
text(hms_cca, display = "species", col = "blue", cex = 1, font = 2)
text(hms_cca, display = "bp", col = "red", font = 2, cex = 1.2)
```



Because the environmental variables explain almost none of the overall symptom variance, the algorithm cannot use them to pull the symptoms apart. Thus, all sixteen symptom variables (in blue) are collapsed into a single, unreadable dot at the origin. This feels like a stark contrast to our expectations given the significance we found in the previous parts. However, while these variables are indeed significantly correlated with distress, it seems they are not the primary, overarching structural drivers of the specific symptom combinations that students experience.

Part 10 | Conclusion

Given our findings from the three ordination techniques applied across Parts 1–9, NMDS emerged as the most robust and informative method. Because the PHQ–9 and GAD–7 scores

rely on ordinal frequency scales with high rates of joint absences (many “0” responses for severe symptoms), the rank-based approach of NMDS paired with the Bray-Curtis distance metric preserved the topological structure of the data far better than the chi-square distances used in Correspondence Analysis (CA). Furthermore, indirect gradient analysis (NMDS with overlays) proved vastly superior to direct gradient analysis (CCA). The CCA demonstrated that forcing the symptom axes to be linear combinations of environmental variables explained only 1.4% of the variance.

Ultimately, the NMDS approach successfully mapped a latent gradient of psychological distress, effectively separating physical and depressive symptoms from anxiety, while correctly isolating severe clinical manifestations like suicidal ideation and self harm. Overlaying variables like loneliness via GAM surfaces provided critical context, proving that while demographic and lifestyle factors are deeply entangled with the severity of a student’s distress, the overarching architecture of their mental health symptoms is highly complex, non-linear, and driven by latent factors beyond basic survey demographics.